Final Project Design Phase

Computer Science - Week 11

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For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I will definitely use conditionals and classes within my program and expect to get better at going through nested data.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest parts will be the conditionals/ if statements because I feel pretty decent on them and LLMS are allowed to check work. The hardest I think will be having 7 pages in my program because I haven't done too much practice with them.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Working the pages and state functions, I will first implement my data classes to set the structure of my code.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

I will minimize distractions by trying to do the work in as few sessions as possible so I can get in the zone and have all my ideas fresh.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Output

What Exercise are you doing?

Bench | Squat | Deadlift

Bench

Body Weight: ___ Bench Weight: ___

Reps: ___

Squat

Body Weight: ___ Squat Weight: ___

Reps: ___

Deadlift

Body Weight: ___ Deadlift Weight: ___

Reps: ___

Your estimated Bench Max is ___	Your estimated Squat Max is ___	Your estimated deadlift Max is ___

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality	The purpose is to help lifters determine their projected 1 rep max lifts in either bench, squat, or deadlift. Have a general opening page in the draft with 3 buttons named for each lift and depending on what button is pushed, the user will enter their bodyweight, exercise weight, and how many reps they did. They will then be returned to a page with an estimation of how much their projected max is after their inputted numbers are put through a formula and returned as an integer.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	class State: name: str body_weight: float exercise: str current_weight: int current_reps: int history: list[Lift] “Data can be stored to compare how many reps and weight has gone up since last used. (Cool way to track progress)”
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified	class Lift: exercise:str weight:str reps: int

(if they are modified)	max: float

Routes (minimum 7)
 	
The names of your route functions
	def index(state):
	def calculator(state):
	def results(state):
	def formulas(state):
	def comparison(state):
	def leaderboard(state):
	def about(state):

Pages (minimum 5) A brief description of the pages in your site	Home Page -Welcome page, may display past data of used before, a button with start calculation on it. Calulation Page -user inputs data and hits “calculate”. Results Page -user sees their results and can hit “view history” to see past results. History Page -past results of max lifts and reps/weights are shown, future foreshadowed results may be shown as well. Information Page -Information about how to improve on certain lifts will be provided and maybe average lifts for certain body weights, age, and gender may be included as well.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter	The user will enter their name, body weight, reps, and exercise weight into their respective textboxes SelectBox will be used to select the type of lift being calculated The textBoxs will be stored into the state function and used to help calculate the 1 rep max. The select box will do the same but by having it as a

Documentation.	select box it will be more convenient for the user as there are only 3 options.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	If bench is selected a higher fatigue integer will be used in the calculation because arms fatigue faster than legs. This just helps with making the estimations as accurate as possible. The same will be done with Squat and Deadlift but lower fatigue values will be used since they are leg exercises. ex.) If exercise == “bench”: formula * 2.5
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	The loop will append a list of old lifts for that type of exercise ex.) bench, to show how progress has changed over time.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Im making a program in python for drafter that will predict a persons 1 rep max lift by having a user input their body weight, exercise weight, and how many reps they did a certain exercise for. However, i need 7 routes, what would be some good names for my route functions.

For a 1 Rep Max Predictor in Drafter, you'll probably want routes that match the different pages/screens of your app. Here are some ideas:

Python

```python
@route
def index(state):
```

- Home page
- Introduces the app

Python

```python
@route
def calculator(state):
```

- Main page where users enter body weight, lift weight, and reps

Python

```python
@route
def results(state):

```

- Displays the predicted 1RM

- Displays the predicted 1RM

Python

```python
@route
def formulas(state):
```

- Explains the formula used (Epley, Brzycki, etc.)

Python

```python
@route
def comparison(state):
```

- Compare different lifts or attempts

Python

```python
@route
def leaderboard(state):
```

- Shows strongest predicted lifters (if you store submissions)

Python

```python
@route
def about(state):
```

- Information about the project and how predictions work

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.